# Product Description

| Product Information | |
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| Product name | Digital Buffer |
| Product number | DB-50-F-A<br>DB-50-R-A |
| Manufacturer | FLIM LABS S.r.l. |
| Registered Address | Via della Farnesina, 3 – 00135 Rome (RM), Italy |




## General Description

The Digital Buffer is a compact electronic module designed for the conversion and distribution of pulse signals into standardized digital signals. It is primarily used in scientific instrumentation, timing, event counting, and fluorescence lifetime imaging microscopy (FLIM) applications, where input digital signals must be adapted for reliable downstream processing.

The product is housed in a compact aluminum enclosure suitable for benchtop laboratory use or integration into portable measurement systems. The metallic enclosure provides mechanical robustness and adequate electromagnetic shielding during operation within the specified limits. The device is equipped with one SMA input and two digital outputs implemented via SMA coaxial connectors. Power is supplied via USB Type-C at 5 V DC.

The Digital Buffer is designed to receive a pulsed digital input signal via SMA, with input voltage levels typically between 1.5 V and 5 V on an internal 50 Ω load. The input signal is internally converted and replicated on two standard LVTTL digital outputs at 2.5 V on 50 Ω, enabling duplication and distribution of the signal to data acquisition devices, timing modules, or other electronic instrumentation. The operating principle is based on active logic-level conversion and distribution of the signal to two electrically independent output channels while maintaining the timing characteristics of the original pulse. The device is intended for integration into existing measurement chains and does not require configuration.

The product is available in two functional variants, which differ only in the input edge detection mode:

- **DB-50-F-A**: optimized for falling-edge input signals
- **DB-50-R-A**: optimized for rising-edge input signals

The variants do not differ in electronic architecture, operating principle, or safety requirements. All modules feature:

- Compact digital buffer module
- One SMA pulse input
- Two SMA digital outputs
- Standard LVTTL 2.5 V outputs on 50 Ω
- USB Type-C 5 V power supply
- Compact aluminum enclosure suitable for system integration
- No user configuration required
- Suitable for signal duplication and distribution
- Power operation compliant with SELV requirements

## Intended Use and Misuse

### Intended Use

The Digital Buffer is intended as a compact electronic module for the conversion and distribution of digital signals by professional users in private research laboratories, companies, and universities. It may be used as a stand-alone unit or integrated into custom measurement systems and instrumentation.

### Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

## Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 0°C to +40°C
- Storage temperature: −20°C to +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases.

Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.